Q1. The photograph shows two rubber tyres.
One is old and worn and the other is new.



old tyre with worn tread

new tyre with deep tread

(a) A car is moving along a road. What force between the tyres and the road makes the car stop? Tick the correct box.

air resistance	☐	friction	☐
gravity	☐	weight	☐

1 mark

(b) The diagram and the table show the stopping distance of a car.



type of road surface	stopping distance, in metres			
	new tyres on a dry road	new tyres on a wet road	old, worn tyres on a dry road	old, worn tyres on a wet road
smooth tarmac	18	19	20	50
rough tarmac	13	18	17	23
concrete	12	17	16	21

(i) What happens to the stopping distance when a road gets wet?

……

1 mark

(ii) Why does the stopping distance change when a road gets wet?

……

……

1 mark

(iii) What happens to the stopping distance as tyres get old and worn?

……

1 mark

(iv) What is the safest type of road surface in the table?

……

1 mark

Maximum 5 marks

Q1.

(a) friction

if more than one box is ticked, award no mark

1 (L3)

(b) (i) it gets longer

accept 'it is more'

1 (L3)

(ii) any **one** from

- there is less friction **or** grip
- the road is slippery

accept 'water is a lubricant'

1 (L4)

(iii) it gets longer

accept 'it is more'

1 (L3)

(iv) concrete

1 (L4)

[5]